

AI-Powered Conversational Assistants for Environmental Management Policy in India

Abstract

Background. Conversational artificial intelligence-this including chatbots, virtual assistants, and large language model (LLM)-based dialogue systems-is being used more and more in Indian public administration, although there has not yet been a systematic examination of its application to environmental management and policy. **Objective.** This scoping review aims to identify what is currently known regarding the use of, and potential use of, the benefits, limitations, risks, and research gaps connected with chatbots and conversational AI acting as assistants in the area of environmental management and policy in India, making use of relevant evidence from India's e-governance sector and from international research on AI in climate and policy areas where direct evidence from India is lacking. **Methods.** The review adopts the Arksey and O'Malley (2005) scoping study framework as improved by Levac, Colquhoun and O'Brien (2010), with reference to the Joanna Briggs Institute (JBI) guidelines and the PRISMA-ScR reporting standards (Tricco et al., 2018), with these adapted to suit a preliminary rather than a final review. The evidence was gathered through a single reviewer conducting a broad search of both scholarly and grey literature. The major bibliographic databases such as Scopus, Web of Science, IEEE Xplore, ACM Digital Library, ScienceDirect, and SpringerLink were not directly searched; this is regarded as the main methodological limitation of the review rather than a minor one. **Results.** There is very little direct evidence from India concerning the use of chatbots in environmental policy. One narrow Indian system with verifiable bibliographic support was found-a drought advisory chatbot developed by the International Water Management Institute in collaboration with ICAR-CRIDA (IWMI & ICAR-CRIDA, 2025). In addition, there is a much larger amount of related evidence: India's general e-governance and grievance redressal chatbot ecosystem, its multilingual digital public infrastructure initiative (Bhashini/AI4Bharat), its national Responsible AI governance documents (NITI Aayog, 2021), and the international literature on retrieval-augmented generation (RAG) and answer faithfulness in climate and policy document domains (Thulke et al., 2024, 2025; Vaghefi et al., 2023; Juhasz et al., 2024). **Conclusions.** The evidence indicates that it is both technically and institutionally feasible for chatbots to serve as information-access and administrative assistance interfaces in Indian environmental governance, but not for them to function as autonomous decision-makers. However, in the Indian context, the reliability of the system with regard to retrieval (faithfulness), its multilingual performance, and the issue of accountability for an environment-specific system have not been adequately evaluated. A conceptual framework and a small number of research priorities are suggested, with these clearly separated from the evidence itself.

Keywords

chatbot; conversational AI; large language models; retrieval-augmented generation; environmental governance; environmental policy; India; e-governance; scoping review; AI accountability

1. Introduction

1.1 Background

Conversational AI has evolved from being based on rules and serving only specific areas to now involving large language model systems that can handle open-domain dialogue, document grounding, and multi-turn reasoning. In several jurisdictions, governments have started to use such systems when providing services to citizens, and India has taken an active role for this development. The Ministry of Electronics and Information Technology in India has organized public innovation challenges that are specifically designed to promote multilingual AI service delivery (AI4Bharat, n.d.), and the relevant policies of the ministry describe a growing number of sector-specific and citizen-oriented conversational systems, as discussed in Section 3.

1.2 The Context of Environmental Governance in India

The system of environmental governance in India is based on a multilayered legal framework, comprising at the national level the Ministry of Environment, Forest and Climate Change (MoEFCC); the Central Pollution Control Board (CPCB) as the main technical and regulatory body, set up in 1974 pursuant to the Water (Prevention and Control of Pollution) Act (CPCB, n.d.); and at the sub-national level, State Pollution Control Boards (SPCBs) in the states and Pollution Control Committees (PCCs) in the Union Territories, which are responsible for setting pollution standards, running monitoring networks, and granting and enforcing industrial consents (CPCB, n.d.). Environmental clearances are now handled via PARIVESH, a single-window digital system developed as part of the Digital India initiative to manage green clearances and keep an eye on compliance across the country (MoEFCC/National Informatics Centre, n.d.). This digital, multi-tiered and document-intensive regulatory system is the type of environment in which conversational, retrieval-based AI assistance has previously been suggested (Section 1.3). However, to the reviewer's knowledge, no academic study has specifically examined whether that suggestion is appropriate within India's particular institutional setup.

1.3 Conversational AI and Environmental Policy

The move from rule-based conversational agents to those based on large language models has led to a well-documented kind of failure known as hallucination, whereby the generated text sounds fluent yet is not based on evidence that can be verified. Retrieval-augmented generation (RAG) - which involves basing the generated text on retrieved external documents rather than depending entirely on the model's parametric knowledge - has become the main solution that is proposed (Juhasz et al., 2024). However, retrieval alone does not ensure that the output stays faithful to the material that was retrieved, as Thulke et al. (2025) demonstrate: a response can be factually accurate with regard to general world knowledge and yet still be unfaithful to the specific passages that were retrieved. This difference is most important in fields like environmental and climate policy, since in these areas standards are versioned, regulations change, and giving incorrect answers can have real regulatory or compliance implications.

1.4 The rationale and the research questions

There are three structural characteristics which make India a special case when considering this question. Firstly, a dynamic digital governance system is already using citizen-service chatbots on a large scale. Secondly, in order to deal with the extreme linguistic diversity, the Bhashini national language initiative - as this review finds - includes all of the 22 languages listed in India's Eighth Schedule (AI4Bharat, n.d.). Thirdly, although the environmental regulatory framework is fragmented it is at the same time becoming more digital, and as a result generates precisely the kind of extensive, technical and frequently updated document collection that systems based on RAG are intended

to handle. Since the relevant literature is broad and highly diverse involving fields like computer science, public administration and environmental policy - and since this particular area of intersection is still in its early stages of development, a scoping review is the suitable approach (Arksey & O'Malley, 2005; Levac et al., 2010). The aim here is to determine the extent and type of the evidence and to identify any gaps, not to combine the effect sizes from a number of comparable studies, since the existing literature does not yet allow for that.

Primary Question: What is at present known regarding the use of chatbots and conversational AI as assistants in environmental management and policy in India, including their potential uses, benefits, limitations, risks, and the existing research gaps?

Sub-questions:

- What chatbots or conversational AI systems have been developed, tested, or suggested for environmental or related purposes (such as e-governance, disaster management, water or agriculture) in India, and what is known about their design?
- What are the benefits and risks reported worldwide of applying LLM/RAG systems to the field of policy documents, and to what extent are these applicable to environmental governance in India?
- What is there to know regarding the multilingual performance and accessibility of such systems in the Indian context?
- What kinds of accountability and human supervision are suggested or mandated for AI systems that are operating within an Indian regulatory environment?

2. Methods

2.1 Scoping Review Approach

The review is based on the five-stage framework developed by Arksey and O'Malley in 2005 and as improved by Levac, Colquhoun and O'Brien in 2010, with reference to the JBI guidelines on methodological issues for scoping reviews, and is generally in accordance with PRISMA-ScR (Tricco et al., 2018). It must be understood as an initial, single-reviewer draft of a scoping review and not as a fully completed systematic scoping review: the complete multi-database search specified below was not carried out, for the practical reasons given in Section 2.4 and Section 7.

2.2 PCC Framework

Population: Citizens, environmental managers, policymakers, regulators (MoEFCC/CPCB/SPCBs), local authorities, and researchers.

Concept: Chatbots, Conversational AI, and LLM based assistants used for environmental information access, policy interpretation, decision support, citizen engagement, and compliance assistance.

Context: India and Indian environmental governance primarily, with international literature included where it directly informs applicability (RAG faithfulness research, comparable e-governance chatbot studies).

2.3 Eligibility Criteria

The following are included: chatbots, conversational AI, and dialogue systems based on large language models that are used or have been explicitly suggested for environmental management, for environmental or climate policy, or

for other areas closely related to public administration (such as e-governance, grievance redressal, and disaster response) where transferable evidence is established; a primary focus should be on systems based in India, with international systems also included if they provide generally applicable technical or governance insights; sources must be peer-reviewed journal or conference papers, arXiv preprints (those marked as unreviewed), or government and institutional reports; the sources should date from 2018 to 2026, with a preference given to those from 2020 to 2026; and the sources must be in English.

Excluded are non-conversational environmental monitoring tools which have no dialogue interface; generic customer-service bots that have no relevance to environmental or governance issues; generic papers that discuss LLM capabilities without focusing on a specific domain; promotional material containing no verifiable technical content; and non-English sources that cannot be independently verified by the reviewer.

2.4 Search Strategy

Because of the limited amount of time available for this assignment, the evidence upon which this draft is based was obtained through an initial, broad search of both scholarly and grey literature rather than by carrying out a fully completed systematic database search. The search was conducted on Google Scholar, arXiv, and ResearchGate, together with a direct review of the documents from government and institutional sources (MoEFCC, CPCB, PARIVESH, NITI Aayog, MeitY/IndiaAI, IWMI, AI4Bharat). In this draft, Scopus, Web of Science, IEEE Xplore, ACM Digital Library, ScienceDirect, and SpringerLink were not searched directly for the same reason - namely the time constraint. The top priority in the next stage should be to perform a complete search of these databases, combining a search block on AI/chatbots, a search block on environmental issues, and a search block on India, before this work can be considered a completed scoping review.

2.5 The selection of studies and the charting of the data

The screening and selection of the studies in accordance with the eligibility criteria set out in Section 2.3 were carried out by one reviewer alone, without carrying out independent dual screening or without conducting a documented check of inter-rater reliability. This represents a significant limitation in comparison with standard procedures in the context of a scoping review and is mentioned here as well as in Section 7 instead of being presented as a fully completed methodology. A structured charting form was employed in order to extract, where provided, the source, year, country, type of evidence, application, environmental domain, policy relevance, target users, key finding, and the evidence strength for each of the included sources (see Table 1).

2.6 Synthesis Approach

The synthesis is narrative and thematically based rather than statistical, in accordance with the conventions of a scoping review and taking into account the heterogeneity of the sources identified. The evidence is clearly divided into three levels in order to prevent presenting evidence from adjacent or international sources as if it directly showed that an Indian environmental-policy chatbot works:

- Tier 1 - Evidence from India's natural environment: systems which are based on or directly serve Indian environmental data or environmental decision-making.
- Tier 2 - evidence that is adjacent to India: this includes e-governance initiatives based in India, systems for redressing grievances, multilingual infrastructure, and evidence relating to AI governance, all of which are not specific to the environment but rather pertain directly to feasibility and design.

- Tier 3 - Evidence that is transferable internationally: this consists of non-Indian research, mainly concerned with climate-policy RAG and faithfulness, and indicates the kind of situation an Indian system might plausibly encounter, even though it does not itself represent Indian evidence.

3. Results

3.1 Evidence Landscape

Since a complete search across multiple databases was not carried out, it is not possible at this stage to produce a numeric PRISMA-ScR flow diagram with identifiable, screening, and inclusion figures that could be defended, and therefore no such diagram is provided. As a result of the preliminary search procedure outlined in Section 2.4, 12 sources were deemed to be sufficiently relevant and verifiable to be kept, covering the three evidence levels defined in Section 2.6. The evidence map that is presented here is illustrative and by no means exhaustive. During the course of the search, several other grey literature sources on Indian e-governance chatbots were found but are not individually cited in this case because the authors could not be reliably verified within the time allowed (see Section 7). One other potential source - a technical document derived from a hackathon that suggested a chatbot for the Central Ground Water Board's INGRES groundwater database - was examined and excluded from the final evidence map since the bibliographic details (including the full author list and formal peer-review status) could not be independently confirmed against the published article. This source is not included in Table 1 nor is it cited anywhere else in this review.

Source	Country	Evidence Type	Application	Environmental Relevance	Policy Relevance	Key Finding	Evidence Strength	Tier
SukhaRakshak AI (IWMI & ICAR-CRIDA, 2025)	India	Institutional pilot report	Multilingual drought-advisory chatbot (Gemini-based RAG), outputs tiered by user role	Direct	Direct - built for farmers, extension officers, and district drought managers	Piloted from mid 2025 onward with user consultation workshops; advisories tiered by role; not yet independently peer reviewed	Low - moderate	1
National Consumer Helpline AI	India	Government case study	NLP/ML enabled chatbot	Indirect	Indirect	Government self	Moderate	2

integration (IndiaAI, 2025)			with workflow automation			reporting of a chatbot assisted consumer grievance routing system		
NITI Aayog Frontier Tech, "Algorithms of Accountability" (NITI Aayog, 2025)	India	Government case compilation	Municipal AI enabled grievance chatbots	Indirect	Indirect - moderate	Documents multi departmental citizen grievance triage via chatbot	Moderate	2
NITI Aayog, <i>Principles for Responsible AI</i> (NITI Aayog, 2021)	India	Government policy document	Cross-sectoral AI governance framework	N/A	Direct	Sets out seven principles: safety and reliability, inclusivity and non discrimination, equality, privacy and security, transparency, accountability, and protection of positive human values	High	2

MeitY/IndiaAI, <i>AI Governance Guidelines</i> (MeitY/IndiaAI, 2025)	India	Government policy document	Cross-sectoral AI governance framework	N/A	Direct	Formalizes accountability, transparency, and human oversight requirements for scalable AI deployment	High	2
Bhashini / AI4Bharat national language mission (AI4Bharat, n.d.)	India	Government/research infrastructure	Multilingual ASR/machine translation/TTS pipeline (22 scheduled languages)	Indirect	Indirect	Provides multilingual infrastructure usable by any citizen facing chatbot, environmental or otherwise	Moderate - high	2
SnehAI instrumental case study (Wang et al., 2022)	India	Peer-reviewed case study	Hinglish chatbot for adolescent sexual/reproductive health education	Not environmental	Indirect (design pattern transfer to sensitive, low-literacy, multilingual topics)	Peer-reviewed evaluation of a taboo topic Indian chatbot using Gibson's affordance theory; high engagement	High (peer-reviewed, <i>Journal of Medical Internet Research</i>)	2

PARIVESH portal description (MoEFCC/NIC, n.d.)	India	Government infrastructure	Single-window digital environmental-clearance platform	Direct	Direct	Establishes the kind of document heavy regulatory corpus a future assistant would need to be grounded in	High (primary government source)	2
ClimateGPT (Thulke et al., 2024)	Germany	Technical/system report	Domain specialized open weight LLM for climate research synthesis, optimised for RAG and multilingual accessibility	Indirect	Direct - designed for climate policy information synthesis	Demonstrates feasibility of a domain specialized, retrieval-oriented climate conversational system	Moderate - high (technical report, released model and benchmark)	3
"Listen to the Context..." (Thulke et al., 2025)	Germany	Empirical evaluation	RAG faithfulness evaluation applied to ClimateGPT	Indirect	Direct - quantifies hallucination/faithfulness risk in policy document RAG	Baseline model showed low faithfulness to retrieved context even with retrieval, improving substantially after targeted	High	3

						fine-tuning		
ChatClimate (Vaghefi et al., 2023)	Switzerland	Peer-reviewed system/evaluation paper	RAG-grounded conversational AI on IPCC reports	Indirect - direct (policy-adjacent scientific grounding)	Indirect	Demonstrates RAG grounding of climate science Q&A in authoritative assessment reports	Moderate - high (peer-reviewed, <i>Communications Earth & Environment</i>) 2	3
Responsible RAG for Climate Decision Making (Juhasz et al., 2024)	United Kingdom	Empirical evaluation	RAG evaluated against climate policy documents with expert faithfulness annotations	Indirect	Direct closest international analogue to an "environmental policy assistant"	Finds a non trivial share of outputs judged not faithful even by expert annotators, who also noted annotation itself is noisy	High (expert-annotated evaluation dataset)	3

Table 1. Evidence map

3.2 Indian Evidence

The only specific example relating to India and the environment at Tier 1 is a single, very narrow and unassessed case. SukhaRakshak AI is a chatbot for providing drought advice based on Gemini and using RAG technology, and it was developed by the International Water Management Institute in collaboration with ICAR's Central Research Institute for Dryland Agriculture. This system is said to adjust the kind of output it provides according to the user's role, offering farmers brief and practical advice, giving extension officers more technical information, and providing district drought managers with a summary type of information (IWMI & ICAR-CRIDA, 2025). It was launched as a result of user consultation workshops that started in mid-2025, and to date it has not published any independently peer-reviewed evaluation data. A second case study - a suggested conversational interface for the Central Ground Water Board's INGRES groundwater-resource database, which was originally proposed as a problem statement in the Smart India Hackathon 2025 - was found during the search but is not included in the evidence base of this review since the bibliographic information in the underlying technical document could not be independently verified

(see Section 7). The fact that this case has been excluded is itself significant: it means that Tier 1 now has only one clearly identifiable Indian system, highlighting just how scarce direct, environment-specific evidence is in this area.

There is a significant amount of evidence relating to India (Tier 2) available outside the environmental field. The overall e-governance and chatbot system for handling complaints in India-as shown in the case of the government's National Consumer Helpline integration (IndiaAI, 2025) and NITI Aayog's records of municipal complaint chatbots (NITI Aayog, 2025)-represents the largest group of examples of India-specific deployments, even though it is mostly general-purpose and not focused on the environment. It is still relevant since environmental complaints such as reports of illegal dumping, objections to industrial pollution, or complaints about trees being felled could reasonably pass through some of the same municipal or departmental channels. That said, none of the sources identified in this search specifically assesses the environmental sub-category, so this link is considered to be inferred rather than directly supported. India's Responsible AI governance documents (NITI Aayog, 2021; MeitY/IndiaAI, 2025) and its multilingual digital infrastructure initiative (AI4Bharat, n.d.) are directly pertinent to the design and governance of an environmental assistant, even though they do not themselves include a study of such an assistant. The Snehai case study (Wang et al., 2022), although not concerned with the environment, provides the clearest example to date of a rigorous, peer-reviewed evaluation of a multilingual chatbot that deals with a sensitive topic and is therefore included here as a methodological example.

3.3 International Evidence Relevant to India

The most substantial evidence presented in this review is of an international (Tier 3) nature and relates to RAG and answer faithfulness in the area of climate policy. ClimateGPT (Thulke et al., 2024) and ChatClimate (Vaghefi et al., 2023) show that it is technically possible to develop domain-specialised, retrieval-grounded conversational systems for climate information, and the fact that ChatClimate has been peer-reviewed in *Communications Earth & Environment* should be noted. Juhasz et al. (2024), who assessed RAG against actual climate-policy legislative text using expert annotations of faithfulness, is the international example most similar to an "environmental-policy assistant". Thulke et al. (2025) offer the most thorough quantitative evidence in the entire review regarding how far even a well-resourced, retrieval-grounded system can fall short of being faithful to the sources it has retrieved. None of this evidence was based on an Indian corpus and it must not be interpreted as demonstrating how an Indian-language system dealing with India's regulatory texts would perform; the evidence is used throughout as transferable material to motivate the evaluation in the Indian context, not as a replacement for such an evaluation.

3.4 Main Application Areas

The same four application areas are found throughout. The area of information access-entailing the transformation of dense regulatory or scientific texts into conversational question-and-answer format-is the one with the most substantial evidence, as shown by ChatClimate and ClimateGPT (Vaghefi et al., 2023; Thulke et al., 2024), and has been proposed but not yet assessed in the case of SukhaRakshak AI in India. Decision support is present only in the advisory system of SukhaRakshak AI, which is the only example from India to use a role-tiered approach, and this aspect has not been evaluated (IWMI & ICAR-CRIDA, 2025). Citizen engagement represents the largest group of evidence specific to India, although it relates generally to e-governance and not to environmental matters (IndiaAI, 2025; NITI Aayog, 2025). Policy interpretation is by far the weakest of the four: none of the sources in this search assess a chatbot's capacity to correctly interpret Indian environmental regulations, for instance the provisions of the Environment (Protection) Act or the PARIVESH consent conditions.

3.5 Benefits and Opportunities

All the evidence shows a clear and focused value proposition - that is, to alleviate the difficulty faced by citizens, public officials, and small industrial operators in dealing with a fragmented and multi-tier regulatory system by turning large volumes of documentation into a conversational interface. This is in line with PARIVESH's own stated reason for being a single-window facilitation tool (MoEFCC/NIC, n.d.). India actually has certain advantages which are not shared by the jurisdictions that produced most of the Tier 3 evidence: a well-established culture of deploying e-governance chatbots, a digital public infrastructure that supports all 22 scheduled languages (AI4Bharat, n.d.), and a clear national policy framework for responsible AI (NITI Aayog, 2021; MeitY/IndiaAI, 2025). Yet these are advantages in terms of building such a system; none of the sources examined shows that they have actually led to a fully functioning and evaluated one.

3.6 Risks and Limitations

The risk best supported by evidence, drawn entirely from Tier 3, is failure of answer faithfulness despite retrieval. Thulke et al. (2025) found that even a domain-specialised model showed low faithfulness to retrieved passages at baseline, and that a response can appear factually correct in general terms while still misrepresenting the specific source it was supposed to be grounded in - a failure mode the authors describe as especially dangerous because it looks trustworthy. Juhasz et al. (2024) independently found a non-trivial share of outputs judged unfaithful even by expert annotators, who also cautioned that human faithfulness judgments themselves are noisy. No Indian-corpus replication of either finding was identified in this search. Multilingual accuracy for environmental terminology, which mixes legal, scientific, and vernacular registers, is an infrastructural strength but an evaluation gap: no source identified in this review reports accuracy or usability metrics for an environmental chatbot operating in a regional Indian language. Accountability for AI-generated environmental compliance guidance - who is responsible if a chatbot gives an industry operator incorrect guidance that is later relied upon - is not addressed by any source located, including India's cross-sectoral governance documents (NITI Aayog, 2021; MeitY/IndiaAI, 2025).

4. Discussion

4.1 What the Evidence Actually Supports

Put together, the evidence leads to a clear and well-defined conclusion regarding the technical and institutional requirements needed for an Indian environmental-policy chatbot: these requirements do exist, except for the specific way in which they are applied in the environmental area. The requirements referred to include RAG methods (Juhasz et al., 2024; Thulke et al., 2025), multilingual infrastructure (AI4Bharat, n.d.), and experience in building bots for the citizen sector (IndiaAI, 2025; NITI Aayog, 2025). This conclusion concerns the feasibility of such a bot, not its performance-the evidence given does not show what level of performance the model would achieve when dealing with regulatory text that is specific to India and written in India-specific languages, in terms of the degree of faithfulness needed to assist with compliance with any particular regulation.

4.2 Relevance to Indian Environmental Governance

If the system is properly constructed and assessed, its potential benefit would be administrative rather than substantive-that is, it would serve to reduce the difficulties involved in dealing with the fragmented regulatory system (the CPCB and the state- and Union Territory-level SPCB/PCC system, as mentioned in Section 1.2; PARIVESH)-rather than replacing the judgment carried out within that system. This interpretation is based on the

structure of the evidence rather than on any actual outcome, since none of the studies identified in this search have measured whether having a conversational interface over PARIVESH-type data actually leads to a reduction in administrative burden in practice. The comparative risks in India are just as distinct as its advantages: the regulatory fragmentation that exists across so many state- and UT-level jurisdictions, the gap in accessibility between rural and urban areas which the Bhashini-type infrastructure only partly covers, and, as far as this review goes, no example of an Indian environmental regulator having published an audit of the system's faithfulness or hallucinations.

4.3 Reliability, Hallucination and RAG

For a policy-grounded assistant two separate properties are important, and the relevant literature regards them as such: factuality, which refers to whether an answer is generally true, and faithfulness, which means that the answer is actually supported by the specific evidence that has been retrieved for it (Thulke et al., 2025). Faithfulness is the more difficult and important of the two properties when it comes to use in the environmental policy area. This is because regulations change, standards have different versions, and a failure in faithfulness could involve confidently citing a withdrawn pollution standard or a revoked notification as if it were current—an error that is more serious than a clearly wrong answer since it is fluent and seems well supported (Thulke et al., 2025). The Tier 3 evidence in Section 3.6 should be understood as a reasonable upper limit on the current level of capability, not as a guarantee of how an Indian-corpus system would perform: the size of the corpus, the quality of retrieval, and the structure of the documents are all different, and no replication using an Indian corpus was found.

4.4 Multilingual and Accessibility Issues

Compared to most of the jurisdictions featured in the Tier 3 evidence, India's most evident infrastructural advantage is the availability of Bhashini and AI4Bharat, which offer translation, speech, and text facilities for all of India's 22 scheduled languages (AI4Bharat, n.d.). It should be stressed that infrastructure is not the same as evaluation. None of the sources cited in this review include accuracy, faithfulness, or usability figures for an environmental chatbot functioning in a regional Indian language, and environmental terminology—which combines legal, scientific, and vernacular registers—is a technically difficult case for precisely the type of multilingual system that Bhashini makes possible.

4.5 Human Oversight and Accountability

The evidence presented here usually leads to the conclusion that conversational AI—whether in the context of India's e-governance sector or in the international climate-related RAG sector—should be regarded as an assistive interface rather than as a body with decision-making powers. The governance terms provided in India's Responsible AI documents—such as transparency, accountability, human oversight, and redress (NITI Aayog, 2021; MeitY/IndiaAI, 2025)—are those against which any future environmental-policy chatbot would have to be evaluated. Yet, as far as this review is aware, the literature specifically on the environmental area has not dealt with the question of who would be accountable if a chatbot gave an operator incorrect compliance advice that was later acted upon.

5. Proposed Conceptual Framework

The conceptual framework set out by the author of this review has been shaped by the literature mentioned above but has not been derived from it. It does not represent an empirical discovery, and there is no evidence within this review to show that a system designed in this way would work well.

The framework consists of six stages which occur in the following sequence: first the User Interface (which can be in text or voice and is available in multiple languages), followed by Conversational AI (this includes intent detection, dialogue management, and the use of the language model), then retrieval from authoritative environmental sources (such as the CPCB and SPCB standards, the PARIVESH filings, and legislation), then Evidence Verification and Citation (involving the validation of the source, checking the version of the document and whether it is still temporally valid), followed by Safety, Uncertainty and Escalation (which involves flagging confidence levels, handing the matter over to a human operator, and keeping an audit log), and finally Human Regulatory Authority, with all final decisions remaining the responsibility of authorized officials.

The main idea behind the design is that the system should help with accessing information and provide administrative support. Any decisions that have a legal or institutional binding effect - such as those concerning consent grants, penalty decisions, and clearance approvals - stay with the duly authorized human actors. This is similar to the retrieval-faithfulness separation described in the international climate-RAG literature (Thulke et al., 2025) and to the stress placed on human oversight in India's Responsible AI principles (NITI Aayog, 2021). The various stages within this framework are not cited individually because the framework as a whole is a proposal and not a collection of separately supported elements.

5.1 Proposed RAG Extension

On the basis of the framework just referred to, a more detailed architecture of retrieval-augmented generation, or RAG, can be developed which is specifically designed for the area of environmental management and policy in India. While a number of traditional RAG implementations place great emphasis on the semantic similarity between a query and the passages retrieved, this limitation is especially significant in this case: in the field of policy, semantic similarity by itself is not enough, because the evidence obtained must also be authoritative, currently in force, jurisdictionally applicable, and relevant to the user's specific situation.

The extension that has been proposed includes a number of steps relating to the selection and verification of evidence within the retrieval layer, in the following order: Query Understanding, Hybrid Retrieval, Policy-Graph Retrieval, Temporal and Authority Filtering, Cross-Encoder Reranking, Evidence Construction, LLM Generation, Faithfulness Verification, and finally either a Citation-Grounded Response or Abstention.

To carry out query understanding, the system first has to determine the user's intent as well as the relevant environmental entities, jurisdiction, policy area, and time frame. For example, in order to answer a question about an emission standard it is necessary to first identify the industry, the pollutant, the state, the regulatory authority, and the applicable period, and this identification must take place before retrieval starts.

What hybrid retrieval involves is not depending entirely on dense semantic retrieval but instead combining vector-based retrieval with lexical and structured policy retrieval. Since regulatory terminology, section numbers, and legal phrasing do not always come up well through semantic similarity on their own, hybrid retrieval is designed to address that recall gap.

Policy-graph retrieval involves representing environmental-policy documents as a graph which captures relationships such as 'issued by', 'applies to', 'amended by', 'supersedes', 'effective from', 'expires on', and 'related to'. This approach enables retrieval to take into account how regulations, amendments, authorities, standards, and jurisdictions relate to one another, rather than considering each document fragment as a separate piece of evidence.

What temporal and authority filtering involves is that the policy assistant must be able to tell the difference between current documents and historical, obsolete, draft, and expired ones. When evidence is retrieved it is checked against metadata including the issuing authority, jurisdiction, publication date and effective date, version, and document status, and the source authority also affects the ranking, official legislation, government notifications, and documents from the CPCB and SPCB being given priority over secondary or informal sources where authoritative material is available. This is important since a document which is semantically relevant is not necessarily the one that is currently in force.

Cross-encoder reranking involves taking the smaller group of candidate passages that have already been retrieved and then re-ranking them using a more computationally intensive relevance model which is able to tell the difference between passages that are generally related to the query and those that actually answer it.

The process of constructing and generating evidence ensures that only the highest quality evidence reaches the language model as part of the generation context, and the response has to maintain explicit connections between its claims and the sources that support them, ideally by including the source title, the name of the issuing authority, information about the document and its version, the effective date where this is applicable, the relevant supporting passage, details on applicability, and an indication of any uncertainty whenever the evidence is insufficient.

Verifying faithfulness and choosing to refrain involves having the response followed by a separate verification step to check whether the claims in the response are actually supported by the evidence retrieved, thus extending the difference between the quality of retrieval and generation faithfulness which has already been discussed in the climate-domain RAG literature (Thulke et al., 2025). If the available evidence is inadequate, contradictory, outdated, or is outside the system's confidence level, the assistant should avoid providing a definite answer and instead refer the matter to the appropriate human authority rather than producing a response that is not supported.

5.2 Research Evaluation of the Proposed Architecture

The greater detail provided by this architecture also implies a specific research programme, because it is not legitimate to assume that any of the additional components will lead to an improvement in performance without first testing them. Each of these components could be assessed by means of experimental comparison with conventional dense RAG, with hybrid retrieval, with and without temporal-validity filtering, with and without source-authority weighting, standard retrieval against policy-graph-enhanced retrieval, reranking against no reranking, and conventional generation against faithfulness-verified generation.

Appropriate evaluation criteria would be retrieval recall and precision, the fraction of authoritative sources that are retrieved, whether the retrieved documents are still valid currently, the accuracy of citations, claim-level faithfulness, the rate at which claims are not supported, the accuracy of abstentions, performance in multilingual retrieval, and assessment by experts in environmental policy rather than relying solely on automated measures.

The proposed framework thus becomes more like a testable research programme than merely an architectural outline. It is still, though, a series of hypotheses requiring empirical validation, and there is no evidence in the current review to indicate that this particular architecture would perform better than conventional RAG; the idea that it does so should not be inferred from the proposal for further research.

6. Research Gaps and Future Research

1. In the search carried out for this review, no peer-reviewed, India-specific empirical assessment of a chatbot used in relation to environmental regulatory text was found. None of the sources examine the accuracy of a chatbot against the standards of the CPCB or the SPCB, PARIVESH filings, or environmental legislation. This represents the most significant gap identified and is a necessary condition for any of the other gaps to be addressed. While the architecture suggested in Section 5.1 is presented as one possible way of organizing such a study, it is the study itself - evaluating any system, whether proposed or not - against actual Indian regulatory text that constitutes the gap, not the architecture.
2. There was no faithfulness or hallucination benchmark found for an Indian environmental corpus. The only faithfulness benchmarks that were located (Thulke et al., 2024, 2025; Juhasz et al., 2024) concern international and English-only materials. Since faithfulness failures are the most serious risk identified in this review, this represents a feasible and high-value first study and is in direct accordance with the faithfulness-verification and abstention stage described in Section 5.1.
3. No study has been found that evaluates the accuracy of environmental-domain conversational AI in the Indian languages. The evidence provided by Bhashini and AI4Bharat is only infrastructural and not concerned with domain evaluation. The combination of legal, scientific, and vernacular registers in environmental terminology means that this represents a non-trivial evaluation problem in itself, and it is specifically mentioned as one of the evaluation measures proposed in Section 5.2.
4. No investigation into the way actual policymakers or regulators in India have used an environmental chatbot was found. The only identifiable Indian Tier 1 example, SukhaRakshak AI, is merely a pre-evaluation pilot. It is just as important to find out whether officials trust, adopt, or misuse such tools as it is to assess technical accuracy, and this point should be included alongside, not in place of, the more technical evaluations suggested in Section 5.2.
5. It was not possible to identify a standardised metric for regulatory faithfulness in a legal or compliance setting. Although general evaluation metrics for RAG are available (Thulke et al., 2025), they have not been adapted for a domain in which the correctness of an answer depends on which version of the regulation is current. The temporal and authority filtering stage suggested in Section 5.1 assumes that such a metric could be developed; yet the task of building and validating it remains unresolved.
6. No accountability or liability framework tailored to AI-generated environmental compliance advice was found. The cross-sectoral AI governance documents of India (NITI Aayog, 2021; MeitY and IndiaAI, 2025) do not specify who should be held responsible if such advice is trusted and is later found to be wrong, since this involves not only a technical issue but also a legal and public administration concern. This absence of a framework cannot be dealt with by any retrieval or generation system, including the one suggested in Section 5.1.
7. No longitudinal evidence of chatbot performance as regulations have changed over time was found. This is due to the general problem of RAG knowledge becoming outdated; none of the Indian deployments mentioned in this search are old enough to have been the subject of longitudinal study. The policy-graph and temporal-filtering components suggested in Section 5.1 are meant to deal with this issue in an architectural sense, but whether or not they actually achieve this remains an open and unexamined question.

When the gaps are considered together with the architecture described in Section 5.1, the obvious next move is to develop a rigorously evaluated benchmark for faithfulness and multilingual accuracy in answering questions about Indian environmental policy, based on a small set of actual CPCB standards and PARIVESH-listed conditions, in a manner methodologically similar to that of Juhasz et al. (2024), and using the component-wise comparisons outlined in Section 5.2 instead of assuming that the proposed architecture has already been validated. Although the

architecture constitutes a possible solution to Gaps 1, 2, 3, 5, and 7, it does not serve as proof that any of these gaps have been closed.

7. Limitations

Because of the limited amount of time that was available for this assignment, the review used Google Scholar, arXiv, and ResearchGate instead of carrying out direct searches of the major bibliographic databases - Scopus, Web of Science, IEEE Xplore, the ACM Digital Library, ScienceDirect, and SpringerLink. This time pressure is the main methodological limitation of the review. Although the resulting body of evidence can serve as a first approximation or initial outline, the search was not complete and the results were not systematically removed of duplicates. The review has to be seen therefore as a preliminary scoping review and not as a fully finished comprehensive evidence synthesis. Moreover, the screening was carried out by only one reviewer, with no independent duplicate screening being performed and no assessment of inter-rater reliability taking place.

A further limitation concerns the nature of the Indian evidence that is available. Several sources provide information about chatbot deployments from the standpoint of the organizations that developed or operated them, for example SukhaRakshak AI, the National Consumer Helpline, and the municipal chatbot initiatives. Although these sources can supply helpful details about the systems and their intended purposes, they do not always amount to independent evidence of effectiveness. Among the materials I examined, I came across no independent and adversarial evaluation of an Indian government environmental or closely related chatbot. Additionally, the search was restricted to content in English, which may have failed to identify relevant work or deployment documents published in Hindi or other Indian languages.

The search also revealed a number of additional grey literature and technical reports, in particular those concerning e-governance chatbots, even though they had not been included in the first selection since the authorship or bibliographic information could not be verified with sufficient confidence. It therefore follows that the fact that so few such systems have been developed cannot be established simply because the small number of Indian examples mentioned here have been cited. The INGRES case illustrates this point. It became clear during the citation check that the case had originated in the problem statement of the Smart India Hackathon 2025 and not from a verified CGWB-endorsed deployment. As the full bibliographic details of the related document could not be independently confirmed, the case was excluded rather than being retained with a classification of weak or uncertain evidence. The only Indian environmental chatbot case that can be directly identified and remains in Tier 1 is SukhaRakshak AI. This should primarily be regarded as a reflection of the limited amount of verifiable evidence that is available, not as proof that no other Indian systems exist.

As there was only a limited quantity of direct evidence from India, the review had to make use to some extent of evidence from other areas and the findings of international studies. The reason for including these sources was to gain an understanding of the technical and governance problems that an environmental-policy assistant in India would face, but applying this evidence to the Indian situation involves making inferences and therefore it cannot be regarded as direct proof of its effectiveness in that case.

Above all, the RAG architecture outlined in Section 5.1 and the evaluation programme described in Section 5.2 must be clearly distinguished from the evidence synthesis. The different components - such as policy-graph retrieval, temporal and authority filtering, cross-encoder reranking, and faithfulness verification with abstention - have been proposed on the basis of the problems identified in this review and relevant international RAG research (Thulke et al., 2025; Juhasz et al., 2024). Yet they have not been implemented or empirically evaluated in this study;

the architecture suggested should therefore be regarded as a guide for future research rather than as a validated solution. It is included in order to convert the research gaps identified into testable research questions, not with the aim of supporting the empirical claims of the review.

8. Conclusion

The available evidence suggests that conversational AI could be used as an assistive interface for access to environmental information and for administrative support in India, although there is still a lack of evidence showing that such systems can be reliably and have been properly evaluated in the context of Indian environmental policy. The only Indian environmental chatbot case that can be clearly identified and verified in this review is a single, narrow and unevaluated institutional pilot, together with a broader set of related e-governance examples and a more thorough international body of research on RAG faithfulness in the climate-policy area. None of this provides direct proof that a chatbot can be relied upon to correctly interpret Indian environmental regulations in the Indian languages at the level of reliability needed by the field. The most clear-cut normative conclusion drawn from both the Indian and international evidence is that these systems should operate under human supervision as tools for accessing information, not as independent decision-makers. The advanced RAG architecture and the evaluation programme proposed in Sections 5.1 and 5.2 are presented as a possible means of addressing that evidence gap, not as a replacement for doing so; they have not yet been tested. The most valuable step for future research should be a rigorous evaluation of a retrieval-based, multilingual environmental-policy assistant using authoritative Indian environmental documents — not the development of another unevaluated prototype.

References

- Arksey, H., & O'Malley, L. (2005). Scoping studies: towards a methodological framework. *International Journal of Social Research Methodology*, 8(1), 19 - 32.
<https://www.tandfonline.com/doi/full/10.1080/1364557032000119616#d1e180>
- Levac, D., Colquhoun, H., & O'Brien, K. K. (2010). Scoping studies: advancing the methodology. *Implementation Science*, 5, 69. <https://link.springer.com/article/10.1186/1748-5908-5-69>
- Tricco, A. C., Lillie, E., Zarin, W., et al. (2018). PRISMA extension for scoping reviews (PRISMA-ScR): checklist and explanation. *Annals of Internal Medicine*, 169(7), 467 - 473.
<https://www.acpjournals.org/doi/10.7326/M18-0850>
- Peters, M. D. J., Godfrey, C., McInerney, P., Munn, Z., Tricco, A. C., & Khalil, H. (2020). Chapter 11: Scoping reviews. In E. Aromataris & Z. Munn (Eds.), *JBIM Manual for Evidence Synthesis*. JBI.
<https://doi.org/10.46658/JBIMES-20-12>
- Thulke, D., Gao, Y., Pelser, P., Brune, R., Jalota, R., Fok, F., Ramos, M., van Wyk, I., Nasir, A., Goldstein, H., Tragemann, T., Nguyen, K., Fowler, A., Stanco, A., Gabriel, J., Taylor, J., Moro, D., Tsymbalov, E., de Waal, J., Matusov, E., Yaghi, M., Shihadah, M., Ney, H., Dugast, C., Dotan, J., & Erasmus, D. (2024). ClimateGPT: Towards AI synthesizing interdisciplinary research on climate change. <https://arxiv.org/abs/2401.09646>
- Thulke, D., Kemmler, J., Dugast, C., & Ney, H. (2025). Listen to the context: towards faithful large language models for retrieval augmented generation on climate questions. In *Proceedings of the 2nd Workshop on Natural Language Processing Meets Climate Change (ClimateNLP 2025)*. Association for Computational Linguistics.
<https://arxiv.org/abs/2505.15633>

- Vaghefi, S. A., Stambach, D., Muccione, V., Bingler, J., Ni, J., Kraus, M., Allen, S., Colesanti-Senni, C., Wekhof, T., Schimanski, T., Gostlow, G., Yu, T., Wang, Q., Webersinke, N., Huggel, C., & Leippold, M. (2023). ChatClimate: grounding conversational AI in climate science. *Communications Earth & Environment*, 4(1), 480. <https://doi.org/10.1038/s43247-023-01084-x>
- Juhasz, M., Dutia, K., Franks, H., Delahunty, C., Mills, P. F., & Pim, H. (2024). Responsible retrieval augmented generation for climate decision making from documents. <https://arxiv.org/abs/2410.23902>
- Wang, H., Gupta, S., Singhal, A., Muttreja, P., Singh, S., Sharma, P., & Piterova, A. (2022). An artificial intelligence chatbot for young people's sexual and reproductive health in India (SnehAI): instrumental case study. *Journal of Medical Internet Research*, 24(1), e29969. <https://doi.org/10.2196/29969>
- International Water Management Institute & ICAR-Central Research Institute for Dryland Agriculture. (2025). *India to get its first AI-based chatbot to tackle drought (SukhaRakshak AI)*. IWMI. <https://www.iwmi.org/blogs/india-to-get-its-first-ai-based-chatbot-to-tackle-drought/>
- IndiaAI/Ministry of Electronics and Information Technology. (2025). *AI-enabled National Consumer Helpline: transforming grievance redressal in India*. Government of India. <https://indiaai.gov.in/article/ai-enabled-national-consumer-helpline-transforming-grievance-redressal-in-india>
- NITI Aayog, Frontier Tech. (2025). *Algorithms of accountability: AI in e-governance for secure and responsive urban systems*. Government of India. <https://frontiertech.niti.gov.in/story/algorithms-of-accountability-ai-in-e-governance-for-secure-and-responsive-urban-systems/>
- NITI Aayog. (2021). *Responsible AI for All: Approach Document for India, Part 1 - Principles for Responsible AI*. Government of India. <https://www.niti.gov.in/sites/default/files/2021-02/Responsible-AI-22022021.pdf>
- Ministry of Electronics and Information Technology/IndiaAI. (2025). *India AI Governance Guidelines*. Government of India. <https://static.pib.gov.in/WriteReadData/specificdocs/documents/2025/nov/doc2025115685601.pdf>
- Ministry of Environment, Forest and Climate Change/National Informatics Centre. (n.d.). *PARIVESH: Pro-Active and Responsive facilitation by Interactive, Virtuous and Environmental Single-Window Hub*. <https://www.impriindia.com/insights/policy-update/proactive-and-responsive-facilitation-by-interactive-and-virtuous-environmental-single-window-hub-parivesh-2018/>
- AI4Bharat. (n.d.). *AI4Bharat: multilingual language technology for India*. IIT Madras / Digital India Bhashini Division, Ministry of Electronics and Information Technology. Available at <https://ai4bharat.iitm.ac.in/areas/llm/>
- Central Pollution Control Board. (n.d.). *Official structure and functions*. Government of India. Available at <https://cpcb.gov.in/functions/>